



Published in final edited form as:

Am J Prev Med. 2015 November ; 49(5 0 3): S230–S240. doi:10.1016/j.amepre.2015.07.019.

Disease Prevention and Health Promotion:

How Integrative Medicine Fits

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Abstract

As a discipline, preventive medicine has traditionally been described to encompass primary, secondary, and tertiary prevention. The fields of preventive medicine and public health share the objectives of promoting general health, preventing disease, and applying epidemiologic techniques to these goals. This paper discusses a conceptual approach between the overlap and potential synergies of integrative medicine principles and practices with preventive medicine in the context of these levels of prevention, acknowledging the relative deficiency of research on the effectiveness of practice-based integrative care. One goal of integrative medicine is to make the widest array of appropriate options available to patients, ultimately blurring the boundaries between conventional and complementary medicine. Both disciplines should be subject to rigorous scientific inquiry so that interventions that are efficacious and effective are systematically distinguished from those that are not. Furthermore, principles of preventive medicine can be infused into prevalent practices in complementary and integrative medicine, promoting public health in the context of more-responsible practices. The case is made that an integrative preventive approach involves the responsible use of science with responsiveness to the needs of patients that persist when conclusive data are exhausted, providing a framework to make clinical decisions among integrative therapies.

Introduction

The dividing line between preventive medicine and public health practice is far from distinct, as is that between prevention and treatment. The purview of preventive medicine has traditionally been described to encompass primary, secondary, and tertiary prevention in the construct usually attributed to Leavell and Clark.¹ Others have expanded on this construct; quaternary prevention focuses on reducing overmedicalization and protecting patients from unnecessary or excessive invasive interventions,² whereas primordial

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No financial disclosures were reported by the authors of this paper.

prevention focuses on the alteration of societal (i.e., environmental, economic, social, behavioral, cultural) structures that affect disease risk.³

This paper discusses a conceptual approach between the overlap and potential synergies of integrative medicine and preventive medicine in the context of these levels of prevention, and represents an update of a prior paper on this topic commissioned by the then IOM (now National Academy of Medicine), and placed in the public domain.⁴

Integrative Medicine and Nomenclature

Integrative medicine, a concept developed over the past few decades,⁵ refers to the fusion—by various means, and to varying degrees—of conventional medical practice and some of the practices that fall under the complementary and alternative medicine (CAM) rubric.^{5, 6} Integrative medicine thus offers, in theory at least, the opportunity to combine the “best” of the conventional healthcare system and practices and providers commonly considered to be CAM,⁷ and thereby produce better outcomes, measured in terms of symptom relief, functional status, patient satisfaction, and perhaps cost effectiveness.⁸ Integrative medicine is necessarily “holistic” in the sense that somatic, emotional, and spiritual health are considered integral to overall health.⁹ These definitions are inherently problematic; what exactly comprises spiritual health, or whether this is the appropriate realm of the physician, is debated.^{10, 11} Further, integrative medicine advocates are accused of creating a forced dichotomy between an idealized patient-centered biopsychosocial approach¹² incorporating CAM and “good conventional medicine.”¹³ A rationale for integrative medicine depends largely on a rationale for CAM, as CAM tends to be the limiting element in efforts to advance integrative care.

The term CAM is used to describe diverse medical practices not routinely taught in mainstream medical education.⁶ “Alternative” denotes that such practices are defined by what they are not, and that they are exclusive of mainstream health care. “Complementary” implies that these practices are supplemental to conventional health care. The discrepancy in suggesting that such practices are both alternative and complementary to mainstream care has been noted.^{4, 14, 15}

Such challenges to the nomenclature notwithstanding, CAM has been the most widely used academic appellation, its primacy conveyed by its incorporation into the title of the NIH National Center for Complementary and Alternative Medicine, recently renamed to the National Center for Complementary and Integrative Health, acknowledging that pure “alternative medicine” is rare, and that “integrative” better conveys prevalent practice patterns.⁶ Despite institutionalization of this terminology, this broad-based categorization overlooks necessary nuance. “CAM” and “integrative” thus encompass practices and practitioners inside and outside of the mainstream, as well as approaches rooted in historic or cultural contexts, unconventional diagnostics and diagnoses, new and untested approaches, and off-label use of conventional therapies. These terms also encompass therapies and approaches that have historically been embraced by CAM clinicians but are recently becoming mainstream (such as some mind–body therapies and interest in the human microbiome beyond gastrointestinal conditions).^{16–18}

Interest in and use of complementary health approaches has remained constant in recent years in adults¹⁹ and children²⁰ after a rise in use between 1990 and 1997.²¹ One third of the adult population¹⁹ and 12% of children²⁰ have used at least one CAM therapy. The majority of patients seek CAM approaches to complement rather than substitute for conventional care most often for pain and chronic musculoskeletal conditions.²²

Americans spent an estimated \$33.9 billion on CAM services in 2007.²³ The use of CAM is more prevalent among female, better-educated, higher-income populations^{21, 22} with chronic and degenerative conditions.^{20, 24, 25} Predictors of CAM use include a holistic philosophical orientation to health and life, a chronic health condition, environmentalism, feminism, and an interest in spirituality and personal growth psychology.²⁵ Other studies show a relationship to health-promoting lifestyle choices: regular physical activity, infrequent to moderate alcohol consumption, and being a former smoker are associated with CAM use.²⁶ Although research findings vary, common reasons that people choose CAM include dissatisfaction with conventional care; a desire to avoid side effects of conventional medicine and treatments; an interest in and greater knowledge of how nutritional, emotional, and lifestyle factors affect health; and a broader focus on disease prevention and overall health.^{21, 24, 25}

Despite prevalent CAM usage, fewer than 40% of CAM patients disclose this information to their mainstream physicians, indicating an important disconnect between patient preferences and comfort in sharing these views.^{21, 25, 27-30} This salient deficiency in provider–patient communication²⁸⁻³⁰ might reflect mistrust, dissatisfaction with the conventional healthcare system,²⁵ or a response to the perceived receptivity of conventional providers.³¹

Therefore, a case may be made to responsibly guide patients in CAM therapies based on interest and in accordance with scientific evidence. Because this guidance should by no means supplant conventional treatments, an argument for an integrative approach emerges: Patients should ideally receive expert guidance across the availability of treatments that may result in improved health.³¹

Integrative Medicine Across the Prevention Spectrum

As behavioral and lifestyle choices account for the majority of premature mortality in the U.S.,³² targeting these areas can potentially provide the greatest benefit. In 2010, the leading cause of death in the U.S. was tobacco use, which resulted in some 435,000 deaths, or 18.1% of total deaths. Closely following was diet and lack of physical activity, resulting in 400,000 deaths.³²

The following sections discuss the potential for integrative medicine across the prevention spectrum. By and large, the effectiveness of integrative approaches in health promotion or disease prevention is not fully elucidated; data derived from direct tests of integrative care models are promising but preliminary.³³⁻³⁶

Integrative Medicine in Primary Prevention

Among the means to promote lifestyle change is modeling (i.e., being an exemplar of) healthy behavior, notably diet and physical activity. Physicians that practice healthy behaviors tend to emphasize these behaviors in patient care; consequently, patients of these physicians generally receive stronger, more pronounced, and more specific advice regarding lifestyle change.^{37, 38} Physicians who exercise regularly are more likely to counsel their patients to do so; nonsmokers are more likely to emphasize the risks of smoking.³⁹

A number of integrative health organizations encourage members to model healthy lifestyle behaviors, including the Academy of Integrative Health & Medicine⁴⁰ and the American Association of Naturopathic Physicians.⁴¹ Among some integrative health educational institutions, a culture of wellness exists, where healthy food choices are readily (if not exclusively) available and faculty model healthy behaviors.

Furthermore, a number of CAM whole systems consider dietary habits and therapeutic nutrition as a cornerstone of health, including Traditional Chinese Medicine,⁴² Ayurveda,^{43, 44} and naturopathy.^{41, 45} Some dietary guidance is consistent with current mainstream recommendations for chronic disease prevention,⁴⁶ whereas some traditional recommendations conflict.

Challenges (and opportunities) also exist in synergizing primary prevention with integrative healthcare. A sizable proportion of patients oriented toward CAM tend to be skeptical of preventive interventions, especially childhood vaccination.^{47, 48} Anti-vaccine views⁴⁹ and increases in vaccine-preventable illnesses are associated with care from CAM providers.⁵⁰ An evidence-based integrative approach in the context of “holistic prevention,” emphasizing the patient–provider relationship,⁵¹ with a sympathetic understanding of parental concerns can potentially increase immunization rates in parents that would otherwise be mistrustful of more-conventional clinicians,⁴⁷ thus protecting public health in the context of providing care that is responsive to the needs of CAM-oriented patients.

Secondary Prevention and Integrative Medicine

Integrative medicine has the potential to improve rates of screening and uptake of preventive services through an emphasis on a strong therapeutic alliance, prevention, teaching, interprofessional, and holistic care.⁵² Nationally, screening rates for preventive services are considerably lower than ideal⁵³; much of the blame can be placed on lack of emphasis and training in health promotion and disease prevention as well as the burdens of a healthcare system that constrain primary care visits to suboptimal levels.⁵⁴⁻⁵⁶ Abbreviated primary care encounters, coupled with barriers to access, tend to compromise continuity of care as well.⁵⁶

As prevention and population health activities occur in almost all healthcare settings,⁵⁷ clinicians can potentially improve screening rates and utilization of preventive services and enhancing risk-reduction efforts for chronic diseases with strong diet and lifestyle associations, namely cardiovascular disease, diabetes, and certain cancers.⁵⁸ Despite this potential, the authors are not aware of evidence of enhanced screening and preventive services in integrative medicine.

Many CAM approaches have demonstrated promise in treating early disease or risk factors such as improving the lipid profile,⁵⁹ reducing inflammation,⁶⁰ controlling serum glucose, and reducing blood pressure.⁶¹⁻⁶⁵ By using these in combination with comprehensive lifestyle change, mind–body interventions, and mainstream preventive recommendations⁶⁶ with a strong therapeutic alliance, the potential to improve outcomes rationally follows.

In certain instances, an integrative approach can be used to enhance adherence with conventional therapies such as using the nutritional supplement coenzyme Q10 to reduce statin-induced myopathy⁶⁷ (though other studies demonstrate a lack of benefit),⁶⁸ probiotics to reduce antibiotic-associated diarrhea,^{69-70, 71} licorice and its derivatives to potentiate the effects of cortisone⁷² and reduce non-steroidal anti-inflammatory drug–associated gastropathy,⁷³ and a variety of integrative approaches to improve quality of life and adverse effects associated with cancer chemotherapy.⁷⁴

Tertiary Prevention and Integrative Medicine

Many lifestyle programs demonstrate effectiveness for tertiary prevention of cardiometabolic disease.⁷⁵⁻⁷⁷ Though aspects of such programs have now arguably been conventionalized (i.e., diet and lifestyle), the blending of lifestyle, dietary supplements, and mind–body interventions is certainly representative, if not diagnostic, of integrative care.³¹

Integrative healthcare approaches for chronic disease can improve functionality, reduce morbidity, improve quality of life, and directly influence disease processes. The quality of evidence for CAM therapies is mixed for treating chronic conditions with significant public health impact.⁷⁸ Nutritional supplements such as fish oil,⁷⁹ chromium,⁸⁰ alpha-lipoic acid,⁸¹ herbal medicines,⁸² and mind–body techniques⁸³ have been used to treat Type 2 diabetes mellitus. Hyperlipidemia can be treated with therapeutic diets consisting of functional foods,^{84, 85} nutritional supplements, and herbal medicines.⁵⁹ Manual therapies such as massage can be useful for osteoarthritis,⁸⁶ as well as acupuncture,⁸⁷ nutritional, and herbal supplements.⁸⁸ An anti-inflammatory diet,^{60, 89} nutritional supplements, manual therapies, and other CAM therapies have shown promise in the management of rheumatoid arthritis.⁹⁰

The public health impact of obesity and its related sequelae is unparalleled in the U.S., while the prevalence is quickly rising throughout the rest of the world.⁹¹ Integrative medicine has the potential to add to obesity prevention and control efforts by emphasizing nutrition, stress reduction,⁹² and exercise.⁹³ There also tends to be an emphasis on nutrient supplements, although the scientific evidence underlying such recommendations has long been suspect.^{94, 95}

At least 13% of outpatient visits are attributable to medically unexplained symptoms^{96, 97} (also known as somatoform disorders) such as chronic fatigue syndrome, irritable bowel syndrome, fibromyalgia, chronic Lyme disease,⁹⁸ and chronic unexplained pain,⁹⁹ which are often complicated by concurrent psychological distress and strong emotions.^{98, 100} Mainstream care for patients with these conditions is often frustrating, usually resulting in extensive diagnostic workups and significant iatrogenic complication rates.^{96, 101} In one study, a majority of primary care physicians described attitudes toward patients as negative

and dismissing,¹⁰² and another study found substantial discordance between patient and physician treatment goals.¹⁰³

As is true of many health conditions that are poorly understood and often resistant to conventional treatments, medically unexplained conditions often compel patients to seek CAM.^{98,99, 104-108} The holistic nature of integrative care, with an emphasis on mind–body medicine, often results in recommendations incorporating psychological and somatic therapies.^{9, 109, 110}

CAM therapies for pain control vary in demonstrated efficacy, spanning mind–body therapies such as meditation^{111, 112} and biofeedback,¹¹³ to tai chi,¹¹⁴ acupuncture, yoga, hypnosis, chiropractic, nutritional interventions,¹¹⁵ herbal medicines, massage,⁸⁶ or combinations thereof.¹¹⁶ In recent years, a number of whole-practice outcomes studies demonstrate benefit of integrative approaches, particularly in chronic pain,^{33, 35, 36, 117-119} Type 2 diabetes,³⁴ and cardiovascular risk markers.¹²⁰ These findings suggest public health benefits as well as possible cost savings.^{8, 120}

Stress and Mind–Body Medicine

Integrative medicine tends to emphasize the importance of psychological stress and its impact on overall health.⁵² The evidence is robust and broad-based; psychological stress leads to poorer health outcomes—encompassing infectious and chronic disease, morbidity and mortality, and developing illness as well as recovery.¹²¹

Psychological states can also be beneficial; the presence of “positive emotions” has been shown to predict better health and outcomes.¹²²⁻¹²⁴ Personality aspects such as commitment to self, an attitude of concern for the environment, a sense of meaningfulness, and an internal locus of control are all associated with decreased illness in high-stress environments.¹²⁵

Contextual factors and the therapeutic relationship are important factors in the overall effectiveness of a therapy, especially with subjective outcomes such as in chronic pain syndromes,^{126, 127} and perhaps stronger with CAM approaches associated with elaborate rituals and distinct contexts.¹²⁸ There is an ethical imperative to provide therapeutic options that are safe and effective for symptomatic relief, with appropriate informed consent, without endorsing approaches that are unsafe or ineffective.¹²⁹ There is an emerging literature on the psychobiology of the placebo effect, with clinically significant effects demonstrated in a variety of contexts.^{128, 130, 131} Intentional use of placebo in clinical practice is routine¹³² with complex ethical implications.¹²⁸

Integrative medicine offers a framework that incorporates psychoemotional factors as integral to overall health with the resultant emphasis on mind–body therapies.¹³³ These factors are often perceived to be overlooked in conventional clinical practice and medical education,¹³⁴⁻¹³⁷ or challenging to practically address in hurried medical visits resulting from financial constraints of the current health delivery system.⁵⁴⁻⁵⁶

Evidence and Integrative Medicine

As integrative medicine often incorporates approaches outside of mainstream care where evidence is weak or speculative, a systematic method in addressing “unconventional therapies” is warranted.³¹ Where strong evidence supporting a particular approach exists, that should be recommended in preference to others. The more ambiguous it is as to which might be the most appropriate therapeutic choice, the more important it is to consider a hierarchy of evidence, incorporating safety, effectiveness, alternatives, and the evidence supporting each (Tables 1 and 2). For some medically unexplained syndromes, such as fibromyalgia or chronic fatigue syndrome, a definitive therapy does not exist, and the best available treatments are those safe and possibly effective. Integrative medicine expands patient options at this end of the evidence hierarchy, where options are generally most needed. Any therapy that a patient refuses to use is ineffective, regardless of the evidence supporting its use.³¹

A common framework to assess the clinical appropriateness of a particular CAM intervention has been published in multiple venues.^{129, 138, 139} Therapies that are both safe and effective are generally recommended, whereas those that are unsafe and ineffective are avoided and discouraged. Areas where either (but not both) safety or efficacy is questioned should be approached with caution. A rational expectation of benefit, based on weak clinical trial evidence or biological plausibility, may be desirable in cases where more evidence-based treatment options are unavailable or undesirable, or when patient preference drives the consideration of a particular intervention. Table 1 illustrates this decision framework.

The authors³¹ have also developed a similar framework to guide clinical recommendations in the context of indefinite research. The expanded Clinical Applications of Research Evidence construct, in Table 2, highlights the practical, and practice-oriented, implications of this interface. These frameworks serve as guides to systematically assess treatment options; clearly, clinical judgment is much more nuanced. Furthermore, there may be challenges in implementing an evidence-based framework for providers that believe therapeutic choice is intuitive and uncompromisingly individualized.^{140, 141}

This framework suggests that clinical application of “evidence” depends on six considerations: the relative safety of a given intervention; its relative effectiveness; the quality and quantity of the supporting evidence; the availability of other treatment options for the condition; cost/accessibility (including insurance coverage, out-of-pocket expense, practicality, availability of reliable providers); and patient preferences. When a treatment approach is unsafe, ineffective, poorly supported by science, less effective than other options, cost prohibitive, and not uniquely compatible with patient preference, it should never be used. When a treatment is safe, effective, supported decisively by science, better than any other therapeutic option, readily accessible, and preferred by a patient, it should always be used. Challenges occur when options reside in between, such as when the approaches supported by the best science have all been tried, and have all failed. What remains is a treatment that is apparently safe, possibly effective, cost neutral, and desired by the patient, but not definitively supported by the available research evidence.³¹

When evaluating any potential clinical intervention, there is an implicit (or explicit) assessment of risk versus benefit. Efficacy is generally the major component of the benefit assessment (but not the only benefit; for example, psychological benefit can occur even in the absence of other clinical effects), whereas safety concerns are the primary risk (but not the only risk; for example, there is economic harm when using a safe but ineffective intervention).

Conclusions

The overlap of integrative medicine with preventive medicine is noteworthy. At the level of primary prevention, a number of approaches can contribute to health promotion. Minimally, these encompass lifestyle counseling, dietary guidance, stress mitigation techniques, interventions to improve sleep quality, and use of natural products for health promotion. At the level of secondary prevention, approaches such as stress management and lifestyle interventions are germane, as are interventions that facilitate use of conventional therapies for risk attenuation. At the level of tertiary prevention, the full range of complementary health approaches pertain to such goals as pain management, symptom control, stress relief, disease management, and risk reduction.

To some extent, a conventional medical system that has emphasized the diagnosis and treatment of disease with ever-increasing degrees of specialization has marginalized both preventive medicine and the holistic view that is central to integrative medicine. The importance of disease prevention/health promotion is gaining increasing recognition, due in part to economic forces molding the evolution of modern health care.¹⁴²⁻¹⁴⁵

As integrative medicine tends to be philosophically aligned toward environmentalism²⁵ and social justice,¹⁴⁶ as well as being particularly concerned with iatrogenesis (adverse effects of medical treatment),¹⁴⁷ the interface of integrative medicine and primordial and quaternary prevention becomes apparent. Integrative medicine thus offers the promise of more-expansive means to achieve the desired ends of preventive medicine, but also imposes the challenges of assessing evidence across that broader expanse.

With patients increasingly interested in complementary and integrative approaches and conventional practitioners often uninformed or reticent, a system of unintegrated or, worse, disintegrated health care prevails in the U.S. Some conventional physicians actively discourage the use of CAM wholesale, without considering the differences in approaches or practitioners—or the potential value of integrative care. CAM-oriented practitioners may be just as apt to discourage the use of standard preventive interventions of conventional medicine, citing its reliance on pharmaceuticals and invasive procedures, a failure to respect nature, a systemic lack of compassion and patient centeredness, and financial conflicts of interest.^{148, 149} It is noteworthy that conflicts of interest (financial and non-financial) and ethical challenges are prevalent in a number of CAM arenas such as providers profiting from dietary supplement sales and laboratory testing.¹⁵⁰⁻¹⁵² There is real danger here of patients toppling into the divide, with attendant squandering of the potential for disease prevention and health promotion.

The Integrative Medicine in Preventive Medicine Education project was designed to introduce preventive medicine residents to integrative medicine to enhance the education and practice of preventive medicine,¹⁵³ implying a unidirectional positive influence of integrative medicine. This is also an opportune time to encourage a bidirectional exchange of ideas where preventive medicine can enhance integrative medicine. In particular, fundamentals of preventive medicine training and practice—biostatistics, epidemiology, research into causes of disease in population groups, the practice of prevention in clinical medicine, and planning and evaluation of health services¹⁵⁴— can improve aspects of CAM and integrative medicine in such areas as childhood vaccines and encourage the critical evaluation of prevalent practices. Indeed, recent national initiatives to limit pharmaceutical industry influence in medical education (PharmFree),¹⁵⁵ better disclosure of financial conflicts of interest (Sunshine Act),¹⁵⁶ and purging low-value practices from medical specialties (Choosing Wisely)¹⁵⁷ can serve as models to improve CAM and integrative health care. Implementing U.S. Food and Drug Administration good manufacturing practices for dietary supplements can mitigate risk of contamination and poor quality control.¹⁵⁸ Incorporating emerging findings from clinical and neurobiological¹⁵⁹ research of contextual and placebo effects can enhance the delivery¹⁶⁰ and understanding of some CAM approaches. Overall, the application of preventive medicine principles and recent quality improvement initiatives can strengthen integrative medicine in the context of a healthcare system moving to value-based care.¹⁴⁵

Recent outcomes research on models (or “whole systems”) of integrative care^{33-36, 117-119} demonstrate promise and innovation, as well as potential cost savings,^{8, 120} despite a lack of large-scale funding^{161, 162} and methodologic challenges.¹⁶³⁻¹⁶⁵ Practice-based research networks of integrative medicine centers are now adding to the literature on community effectiveness of integrative medicine for chronic pain¹¹⁷ and cancer care.¹⁶⁶ Initiatives in integrative practices in underserved communities¹⁶⁷⁻¹⁶⁹ demonstrate a public health orientation beyond the more-educated and -affluent demographics historically associated with CAM. Furthermore, the Patient-Centered Outcomes Research Institute, established by Congress in the Patient Protection and Affordable Care Act of 2010,¹⁷⁰ is focused on patient-centered comparative clinical effectiveness research and includes explicit stipulations for research on “integrative health practices.”¹⁷⁰ Relevant PCORI-funded research focused on and chronic pain in underserved communities include studies in acupuncture¹⁷¹ and integrative group visits.¹⁷² Some posit that these emerging findings can influence the current healthcare system to be both more value driven and more aligned to integrative principles.¹⁷³

The ultimate goal of integrative medicine should be to make the widest array of appropriate options available to patients. Appropriateness should be predicated on fundamental considerations that pertain equally to conventional and CAM practice: treatment safety and treatment effectiveness. Treatment safety and treatment effectiveness must, in turn, be interpreted in light of the available evidence.

Evaluating the current state of integrative medicine and preventive medicine leads to a number of fundamental research questions that can address essential gaps. The role of integrative medicine in areas germane to primary and secondary prevention needs to better

assessed in terms of rates of uptake of clinical preventive services, as well as the ability to demonstrably improve diet, physical activity, and smoking-cessation efforts. The question of whether integrative clinicians model healthy behavior and influence their patients in comparison to other clinicians is relevant, as well as cost-effectiveness studies of integrative practices in high-priority areas such as pain management and adjunctive cancer care.¹⁷⁴

But even in the absence of evidence, health care is not advanced by failing to adequately treat symptoms, engage patients in a therapeutic alliance, control disease progression, or produce satisfaction.³¹ The simple argument supporting integrative care is that modern medical science and knowledge, despite profound successes, comprises far less than patient need. Integrative care is not a comprehensive solution, but does expand the array of patient options, and can increase the likelihood of success that can be assembled across the stages of prevention.³¹

Acknowledgments

The authors would like to thank Asim A. Jani, MD, MPH for contributing insights and expertise in preparing this manuscript.

The authors are grateful for the funding support from the Health Resources and Services Administration (HRSA) of the U.S. DHHS, and their Integrative Medicine Program, HRSA grant #IMOPH25100. This information or content and conclusions are those of the authors and should not be construed as the official position or policy of, nor should any endorsements be inferred by, the HRSA, DHHS, or U.S. Government.

The project was supported by the HRSA of the U.S. DHHS under grant #UE1HP25094 and from the National Center for Complementary and Integrative Health at the National Institutes of Health under grant #K23AT006703.

References

1. Leavell, H.; Clark, E. Textbook of Preventive Medicine. 3. McGraw-Hill; New York: 1953.
2. Jamouille M. Quaternary prevention, an answer of family doctors to overmedicalization. *Int J Health Policy Manag.* 2015; 4(2):61–64. <http://dx.doi.org/10.15171/ijhpm.2015.24>. [PubMed: 25674569]
3. Porta, MS.; International Epidemiological Association. A dictionary of epidemiology. 5th. Oxford University Press; Oxford ; New York: 2008.
4. Katz, D.; Ali, A. Preventive Medicine, Integrative Medicine, and the Health of the Public; Commissioned paper for IOM of the National Academies. Summit on Integrative Medicine and the Health of the Public; 2009.
5. Jonas WB, Eisenberg D, Hufford D, Crawford C. The evolution of complementary and alternative medicine (CAM) in the USA over the last 20 years. *Forsch Komplementmed.* 2013; 20(1):65–72. <http://dx.doi.org/10.1159/000348284>. [PubMed: 23727764]
6. National Center for Complementary and Integrative Health. Frequently Asked Questions: Name Change. 2014. <https://nccih.nih.gov/news/name-change-faq>
7. Ring M, Brodsky M, Low Dog T, et al. Developing and implementing core competencies for integrative medicine fellowships. *Acad Med.* 2014; 89(3):421–428. <http://dx.doi.org/10.1097/ACM.000000000000148>. [PubMed: 24448047]
8. Herman PM, Poindexter BL, Witt CM, Eisenberg DM. Are complementary therapies and integrative care cost-effective? A systematic review of economic evaluations. *BMJ Open.* 2012; 2(5) <http://dx.doi.org/10.1136/bmjopen-2012-001046>.
9. Goldstein MS, Sutherland C, Jaffe DT, Wilson J. Holistic physicians and family practitioners: similarities, differences and implications for health policy. *Soc Sci Med.* 1988; 26(8):853–861. [http://dx.doi.org/10.1016/0277-9536\(88\)90178-5](http://dx.doi.org/10.1016/0277-9536(88)90178-5). [PubMed: 3375857]

10. Luster L, Hines B. Debate question: should physicians incorporate spirituality into the care of patients? *South Med J*. 2005; 98(12):1242. <http://dx.doi.org/10.1097/01.smj.0000190307.33328.f1>. [PubMed: 16440938]
11. Scheurich N. Reconsidering spirituality and medicine. *Acad Med*. 2003; 78(4):356–360. <http://dx.doi.org/10.1097/00001888-200304000-00005>. [PubMed: 12691963]
12. Smith RC, Fortin AH, Dwamena F, Frankel RM. An evidence-based patient-centered method makes the biopsychosocial model scientific. *Patient Educ Couns*. 2013; 91(3):265–270. <http://dx.doi.org/10.1016/j.pec.2012.12.010>. [PubMed: 23352913]
13. McLachlan JC. Integrative medicine and the point of credulity. *BMJ*. 2010; 341:c6979. <http://dx.doi.org/10.1136/bmj.c6979>. [PubMed: 21147748]
14. Druss BG, Rosenheck RA. Association between use of unconventional therapies and conventional medical services. *JAMA*. 1999; 282(7):651–656. <http://dx.doi.org/10.1001/jama.282.7.651>. [PubMed: 10517718]
15. Katz DL. Conventional medical care and unconventional therapies. *JAMA*. 2000; 283(1):56. author reply 57. <http://dx.doi.org/10.1001/jama.283.1.56>. [PubMed: 10632280]
16. Hur KY, Lee MS. Gut Microbiota and Metabolic Disorders. *Diabetes Metab J*. 2015; 39(3):198–203. <http://dx.doi.org/10.4093/dmj.2015.39.3.198>. [PubMed: 26124989]
17. Giorgetti G, Brandimarte G, Fabiocchi F, et al. Interactions between Innate Immunity, Microbiota, and Probiotics. *J Immunol Res*. 2015; 2015:501361. <http://dx.doi.org/10.1155/2015/501361>. [PubMed: 26090492]
18. Zhou L, Foster JA. Psychobiotics and the gut-brain axis: in the pursuit of happiness. *Neuropsychiatr Dis Treat*. 2015; 11:715–723. [PubMed: 25834446]
19. Clarke TC, Black LI, Stussman BJ, Barnes PM, Nahin RL. Trends in the use of complementary health approaches among adults: United States, 2002–2012. *Natl Health Stat Report*. 2015; (79):1–16.
20. Black LI, Clarke TC, Barnes PM, Stussman BJ, Nahin RL. Use of complementary health approaches among children aged 4–17 years in the United States: national health interview survey, 2007–2012. *Natl Health Stat Report*. 2015; (78):1–19.
21. Eisenberg DM, Davis RB, Ettner SL, et al. Trends in Alternative Medicine Use in the United States, 1990–1997: Results of a Follow-up National Survey. *JAMA*. 1998; 280(18):1569–1575. <http://dx.doi.org/10.1001/jama.280.18.1569>. [PubMed: 9820257]
22. Barnes PM, Bloom B, Nahin RL. Complementary and alternative medicine use among adults and children: United States, 2007. *Natl Health Stat Report*. 2008; (12):1–23. [PubMed: 19361005]
23. Nahin RL, Barnes PM, Stussman BJ, Bloom B. Costs of complementary and alternative medicine (CAM) and frequency of visits to CAM practitioners: United States, 2007. *Natl Health Stat Report*. 2009; (18):1–14. [PubMed: 19771719]
24. Barnes, PM.; Powell-Griner, E.; McFann, K.; Nahin, RL. Complementary and alternative medicine use among adults: United States, 2002; *Adv Data*. 2004. p. 1–19. <http://dx.doi.org/10.1016/j.sigm.2004.07.003>
25. Astin JA. Why patients use alternative medicine: results of a national study. *JAMA*. 1998; 279(19):1548–1553. <http://dx.doi.org/10.1001/jama.279.19.1548>. [PubMed: 9605899]
26. Nahin RL, Dahlhamer JM, Taylor BL, et al. Health behaviors and risk factors in those who use complementary and alternative medicine. *BMC Public Health*. 2007; 7:217. <http://dx.doi.org/10.1186/1471-2458-7-217>. [PubMed: 17723149]
27. Eisenberg DM, Kessler RC, Foster C, Norlock FE, Calkins DR, Delbanco TL. Unconventional medicine in the United States. Prevalence, costs, and patterns of use. *N Engl J Med*. 1993; 328(4):246–252. <http://dx.doi.org/10.1056/NEJM199301283280406>. [PubMed: 8418405]
28. Elder NC, Gillcrist A, Minz R. Use of alternative health care by family practice patients. *Arch Fam Med*. 1997; 6(2):181–184. <http://dx.doi.org/10.1001/archfami.6.2.181>. [PubMed: 9075455]
29. Feldman MK. Patients who seek unorthodox medical treatment. *Minn Med*. 1990; 73(6):19–25.
30. McKee J. Holistic health and the critique of Western medicine. *Soc Sci Med*. 1988; 26(8):775–784. [http://dx.doi.org/10.1016/0277-9536\(88\)90171-2](http://dx.doi.org/10.1016/0277-9536(88)90171-2). [PubMed: 3287633]

31. Katz, D.; Ali, A. Preventive Medicine, Integrative Medicine, and the Health of the Public; Commissioned paper for Institute of Medicine (IOM) of the National Academies. Summit on Integrative Medicine and the Health of the Public; 2009.
32. Mokdad AH, Marks JS, Stroup DF, Gerberding JL. Actual causes of death in the United States, 2000. *JAMA*. 2004; 291(10):1238–1245. <http://dx.doi.org/10.1001/jama.291.10.1238>. [PubMed: 15010446]
33. Johnson JR, Crespin DJ, Griffin KH, Finch MD, Dusek JA. Effects of integrative medicine on pain and anxiety among oncology inpatients. *J Natl Cancer Inst Monogr*. 2014; 2014(50):330–337. <http://dx.doi.org/10.1093/jncimonographs/lgu030>. [PubMed: 25749600]
34. Bradley R, Sherman KJ, Catz S, et al. Adjunctive naturopathic care for type 2 diabetes: patient-reported and clinical outcomes after one year. *BMC Complement Altern Med*. 2012; 12:44. <http://dx.doi.org/10.1186/1472-6882-12-44>. [PubMed: 22512949]
35. Johnson JR, Crespin DJ, Griffin KH, et al. The effectiveness of integrative medicine interventions on pain and anxiety in cardiovascular inpatients: a practice-based research evaluation. *BMC Complement Altern Med*. 2014; 14:486. <http://dx.doi.org/10.1186/1472-6882-14-486>. [PubMed: 25494710]
36. Szczurko O, Cooley K, Busse JW, et al. Naturopathic care for chronic low back pain: a randomized trial. *PLoS One*. 2007; 2(9):e919. <http://dx.doi.org/10.1371/journal.pone.0000919>. [PubMed: 17878954]
37. Oberg EB, Frank E. Physicians' health practices strongly influence patient health practices. *J R Coll Physicians Edinb*. 2009; 39(4):290–291. <http://dx.doi.org/10.4997/JRCPE.2009.422>. [PubMed: 21152462]
38. Maheux B, Pineault R, Lambert J, Beland F, Berthiaume M. Factors influencing physicians' preventive practices. *Am J Prev Med*. 1989; 5(4):201–206. [PubMed: 2765290]
39. Frank E, Kunovich-Frieze T. Physicians' prevention counseling behaviors: current status and future directions. *Prev Med*. 1995; 24(6):543–545. <http://dx.doi.org/10.1006/pmed.1995.1086>. [PubMed: 8610075]
40. Academy of Integrative Health & Medicine. Academy Values. 2015. <http://aihm.org/about/academy-values/>
41. Hough, H.; Dower, C.; O'Neil, E. Profile of a Profession: Naturopathic Practice. University of California, San Francisco, Centre for the Health Professions; San Francisco: 2001.
42. Kaptchuk, T. The Web That Has No Weaver: Understanding Chinese Medicine. 2. McGraw-Hill; 2000.
43. National Center for Complementary and Integrative Health. Ayurvedic Medicine: An Introduction. 2013.
44. Chopra A, Doiphode VV. Ayurvedic medicine. Core concept, therapeutic principles, and current relevance. *Med Clin North Am*. 2002; 86(1):75–89. vii. [http://dx.doi.org/10.1016/S0025-7125\(03\)00073-7](http://dx.doi.org/10.1016/S0025-7125(03)00073-7). [PubMed: 11795092]
45. National Center for Complementary and Integrative Health. Naturopathy: An Introduction. 2012. <https://nccih.nih.gov/health/naturopathy/naturopathyintro.htm>
46. Katz DL, Meller S. Can we say what diet is best for health? *Annu Rev Public Health*. 2014; 35:83–103. <http://dx.doi.org/10.1146/annurev-publhealth-032013-182351>. [PubMed: 24641555]
47. Benin AL, Wisler-Scher DJ, Colson E, Shapiro ED, Holmboe ES. Qualitative analysis of mothers' decision-making about vaccines for infants: the importance of trust. *Pediatrics*. 2006; 117(5): 1532–1541. <http://dx.doi.org/10.1542/peds.2005-1728>. [PubMed: 16651306]
48. Stokley S, Cullen KA, Kennedy A, Bardenheier BH. Adult vaccination coverage levels among users of complementary/alternative medicine - results from the 2002 National Health Interview Survey (NHIS). *BMC Complement Altern Med*. 2008; 8:6. <http://dx.doi.org/10.1186/1472-6882-8-6>. [PubMed: 18294382]
49. Ali A, Calabrese C, Lee R, Salmon D, Zwickey H. Vaccination Attitudes and Education in Naturopathic Medicine Students. *J Altern Complement Med*. 2014; 20(5):A115–A116. <http://dx.doi.org/10.1089/acm.2014.5307.abstract>.
50. Downey L, Tyree PT, Huebner CE, Lafferty WE. Pediatric vaccination and vaccine-preventable disease acquisition: associations with care by complementary and alternative medicine providers.

- Matern Child Health J. 2010; 14(6):922–930. <http://dx.doi.org/10.1007/s10995-009-0519-5>. [PubMed: 19760163]
51. Academic Consortium for Integrative Medicine & Health. About Us. 2015. <http://www.imconsortium.org/about/home.cfm>
 52. Snyderman R, Weil AT. Integrative medicine: bringing medicine back to its roots. Arch Intern Med. 2002; 162(4):395–397. <http://dx.doi.org/10.1001/archinte.162.4.395>. [PubMed: 11863470]
 53. Healthy People 2020. Clinical Preventive Services. 2014. www.healthypeople.gov/2020/leading-health-indicators/2020-lhi-topics/Clinical-Preventive-Services
 54. McGlynn EA, Asch SM, Adams J, et al. The quality of health care delivered to adults in the United States. N Engl J Med. 2003; 348(26):2635–2645. <http://dx.doi.org/10.1056/NEJMsa022615>. [PubMed: 12826639]
 55. Mangione-Smith R, DeCristofaro AH, Setodji CM, et al. The quality of ambulatory care delivered to children in the United States. N Engl J Med. 2007; 357(15):1515–1523. <http://dx.doi.org/10.1056/NEJMsa064637>. [PubMed: 17928599]
 56. Bodenheimer T. Coordinating care--a perilous journey through the health care system. N Engl J Med. 2008; 358(10):1064–1071. <http://dx.doi.org/10.1056/NEJMhpr0706165>. [PubMed: 18322289]
 57. Zenzano T, Allan JD, Bigley MB, et al. The roles of healthcare professionals in implementing clinical prevention and population health. Am J Prev Med. 2011; 40(2):261–267. <http://dx.doi.org/10.1016/j.amepre.2010.10.023>. [PubMed: 21238876]
 58. Ford ES, Bergmann MM, Kroger J, Schienkiewitz A, Weikert C, Boeing H. Healthy living is the best revenge: findings from the European Prospective Investigation Into Cancer and Nutrition-Potsdam study. Arch Intern Med. 2009; 169(15):1355–1362. <http://dx.doi.org/10.1001/archinternmed.2009.237>. [PubMed: 19667296]
 59. Nies LK, Cymbala AA, Kasten SL, Lamprecht DG, Olson KL. Complementary and alternative therapies for the management of dyslipidemia. Ann Pharmacother. 2006; 40(11):1984–1992. <http://dx.doi.org/10.1345/aph.1H040>. [PubMed: 17047144]
 60. Kohatsu W. Integrative Medicine. In: Rakel, DP., editor. The Antiinflammatory Diet. Third. Elsevier; Philadelphia, PA: 2012. p. 795–802.
 61. Lee T, Dugoua JJ. Nutritional supplements and their effect on glucose control. Adv Exp Med Biol. 2012; 771:381–395. [PubMed: 23393691]
 62. Li GQ, Kam A, Wong KH, et al. Herbal medicines for the management of diabetes. Adv Exp Med Biol. 2012; 771:396–413. [PubMed: 23393692]
 63. Houston M. Nutrition and nutraceutical supplements for the treatment of hypertension: part I. J Clin Hypertens (Greenwich). 2013; 15(10):752–757. [PubMed: 24088285]
 64. Yeh GY, Eisenberg DM, Kaptchuk TJ, Phillips RS. Systematic review of herbs and dietary supplements for glycemic control in diabetes. Diabetes Care. 2003; 26(4):1277–1294. <http://dx.doi.org/10.2337/diacare.26.4.1277>. [PubMed: 12663610]
 65. Brook RD, Jackson EA, Giorgini P, McGowan CL. When and How to Recommend "Alternative Approaches" in the Management of High Blood Pressure. Am J Med. 2015; 128(6):567–570. <http://dx.doi.org/10.1016/j.amjmed.2014.12.029>. [PubMed: 25644320]
 66. The Guide to Clinical Preventive Services 2014: Recommendations of the U.S. Preventive Services Task Force. Rockville (MD): 2014.
 67. Skarlovnik A, Janic M, Lunder M, Turk M, Sabovic M. Coenzyme Q10 supplementation decreases statin-related mild-to-moderate muscle symptoms: a randomized clinical study. Med Sci Monit. 2014; 20:2183–2188. <http://dx.doi.org/10.12659/MSM.890777>. [PubMed: 25375075]
 68. Taylor BA, Lorson L, White CM, Thompson PD. A randomized trial of coenzyme Q10 in patients with confirmed statin myopathy. Atherosclerosis. 2015; 238(2):329–335. <http://dx.doi.org/10.1016/j.atherosclerosis.2014.12.016>. [PubMed: 25545331]
 69. Floch MH. Recommendations for probiotic use in humans-a 2014 update. Pharmaceuticals (Basel). 2014; 7(10):999–1007. <http://dx.doi.org/10.3390/ph7100999>. [PubMed: 25310351]
 70. McFarland LV. Meta-analysis of probiotics for the prevention of antibiotic associated diarrhea and the treatment of Clostridium difficile disease. Am J Gastroenterol. 2006; 101(4):812–822. <http://dx.doi.org/10.1111/j.1572-0241.2006.00465.x>. [PubMed: 16635227]

71. Johnston, BC.; Supina, AL.; Ospina, M.; Vohra, S. Probiotics for the prevention of pediatric antibiotic-associated diarrhea; Cochrane Database Syst Rev. 2007. p. CD004827<http://dx.doi.org/10.1002/14651858.cd004827.pub2>
72. Teelucksingh S, Mackie AD, Burt D, McIntyre MA, Brett L, Edwards CR. Potentiation of hydrocortisone activity in skin by glycyrrhetic acid. *Lancet*. 1990; 335(8697):1060–1063. [http://dx.doi.org/10.1016/0140-6736\(90\)92633-S](http://dx.doi.org/10.1016/0140-6736(90)92633-S). [PubMed: 1970371]
73. Russell RI, Morgan RJ, Nelson LM. Studies on the protective effect of deglycyrrhised liquorice against aspirin (ASA) and ASA plus bile acid-induced gastric mucosal damage, and ASA absorption in rats. *Scand J Gastroenterol Suppl*. 1984; 92:97–100. [PubMed: 6588541]
74. Greenlee H, Balneaves LG, Carlson LE, et al. Clinical practice guidelines on the use of integrative therapies as supportive care in patients treated for breast cancer. *J Natl Cancer Inst Monogr*. 2014; 2014(50):346–358. <http://dx.doi.org/10.1093/jncimonographs/lgu041>. [PubMed: 25749602]
75. Ornish D, Scherwitz LW, Billings JH, et al. Intensive lifestyle changes for reversal of coronary heart disease. *JAMA*. 1998; 280(23):2001–2007. <http://dx.doi.org/10.1001/jama.280.23.2001>. [PubMed: 9863851]
76. Knowler WC, Barrett-Connor E, Fowler SE, et al. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *N Engl J Med*. 2002; 346(6):393–403. <http://dx.doi.org/10.1056/NEJMoa012512>. [PubMed: 11832527]
77. de Lorgeril M, Salen P, Martin JL, Monjaud I, Delaye J, Mamelle N. Mediterranean diet, traditional risk factors, and the rate of cardiovascular complications after myocardial infarction: final report of the Lyon Diet Heart Study. *Circulation*. 1999; 99(6):779–785. <http://dx.doi.org/10.1161/01.CIR.99.6.779>. [PubMed: 9989963]
78. Ward BW, Schiller JS. Prevalence of multiple chronic conditions among US adults: estimates from the National Health Interview Survey, 2010. *Prev Chronic Dis*. 2013; 10:E65. <http://dx.doi.org/10.5888/pcd10.120203>. [PubMed: 23618545]
79. Hartweg, J.; Perera, R.; Montori, V.; Dinneen, S.; Neil, HA.; Farmer, A. Omega-3 polyunsaturated fatty acids (PUFA) for type 2 diabetes mellitus; Cochrane Database Syst Rev. 2008. p. CD003205<http://dx.doi.org/10.1002/14651858.cd003205.pub2>
80. Balk EM, Tatsioni A, Lichtenstein AH, Lau J, Pittas AG. Effect of chromium supplementation on glucose metabolism and lipids: a systematic review of randomized controlled trials. *Diabetes Care*. 2007; 30(8):2154–2163. <http://dx.doi.org/10.2337/dc06-0996>. [PubMed: 17519436]
81. Singh U, Jialal I. Alpha-lipoic acid supplementation and diabetes. *Nutr Rev*. 2008; 66(11):646–657. <http://dx.doi.org/10.1111/j.1753-4887.2008.00118.x>. [PubMed: 19019027]
82. Bradley R, Oberg EB, Calabrese C, Standish LJ. Algorithm for complementary and alternative medicine practice and research in type 2 diabetes. *J Altern Complement Med*. 2007; 13(1):159–175. <http://dx.doi.org/10.1089/acm.2006.6207>. [PubMed: 17309390]
83. Kligler B. The role of the optimal healing environment in the care of patients with diabetes mellitus type II. *J Altern Complement Med*. 2004; 10(Suppl 1):S223–229. <http://dx.doi.org/10.1089/1075553042245926>. [PubMed: 15630839]
84. Becker DJ, Gordon RY, Morris PB, et al. Simvastatin vs therapeutic lifestyle changes and supplements: randomized primary prevention trial. *Mayo Clin Proc*. 2008; 83(7):758–764. <http://dx.doi.org/10.4065/83.7.758>. [PubMed: 18613992]
85. Jenkins DJ, Kendall CW, Faulkner DA, et al. Assessment of the longer-term effects of a dietary portfolio of cholesterol-lowering foods in hypercholesterolemia. *Am J Clin Nutr*. 2006; 83(3):582–591. [PubMed: 16522904]
86. Perlman AI, Sabina A, Williams AL, Njike VY, Katz DL. Massage therapy for osteoarthritis of the knee: a randomized controlled trial. *Arch Intern Med*. 2006; 166(22):2533–2538. <http://dx.doi.org/10.1001/archinte.166.22.2533>. [PubMed: 17159021]
87. Berman BM, Lao L, Langenberg P, Lee WL, Gilpin AM, Hochberg MC. Effectiveness of acupuncture as adjunctive therapy in osteoarthritis of the knee: a randomized, controlled trial. *Ann Intern Med*. 2004; 141(12):901–910. <http://dx.doi.org/10.7326/0003-4819-141-12-200412210-00006>. [PubMed: 15611487]
88. Ernst E. Complementary or alternative therapies for osteoarthritis. *Nat Clin Pract Rheumatol*. 2006; 2(2):74–80. <http://dx.doi.org/10.1038/ncprheum0093>. [PubMed: 16932660]

89. Adam O, Beringer C, Kless T, et al. Anti-inflammatory effects of a low arachidonic acid diet and fish oil in patients with rheumatoid arthritis. *Rheumatol Int.* 2003; 23(1):27–36. [PubMed: 12548439]
90. Muller, D. Rheumatoid Arthritis. Integrative Medicine. Third. Rakel, DP., editor. Elsevier; Philadelphia, PA: 2012. p. 456-463.
91. Ng M, Fleming T, Robinson M, et al. Global, regional, and national prevalence of overweight and obesity in children and adults during 1980-2013: a systematic analysis for the Global Burden of Disease Study 2013. *Lancet.* 2014; 384(9945):766–781. [http://dx.doi.org/10.1016/S0140-6736\(14\)60460-8](http://dx.doi.org/10.1016/S0140-6736(14)60460-8). [PubMed: 24880830]
92. O'Reilly GA, Cook L, Spruijt-Metz D, Black DS. Mindfulness-based interventions for obesity-related eating behaviours: a literature review. *Obes Rev.* 2014; 15(6):453–461. <http://dx.doi.org/10.1111/obr.12156>. [PubMed: 24636206]
93. Nicolai, J.; Lupiani, J.; Wolf, A. An Integrative Approach to Obesity. In: Rakel, DP., editor. Integrative Medicine. Third. Elsevier; Philadelphia, PA: 2012. p. 364-375.
94. Vaughan RA, Conn CA, Mermier CM. Effects of commercially available dietary supplements on resting energy expenditure: a brief report. *ISRN Nutr.* 2014; 2014:650264. <http://dx.doi.org/10.1155/2014/650264>. [PubMed: 24967272]
95. Allison DB, Fontaine KR, Heshka S, Mentore JL, Heymsfield SB. Alternative treatments for weight loss: a critical review. *Crit Rev Food Sci Nutr.* 2001; 41(1):1–28. discussion 39-40. <http://dx.doi.org/10.1080/20014091091661>. [PubMed: 11152041]
96. Ring A, Dowrick CF, Humphris GM, Davies J, Salmon P. The somatising effect of clinical consultation: what patients and doctors say and do not say when patients present medically unexplained physical symptoms. *Soc Sci Med.* 2005; 61(7):1505–1515. <http://dx.doi.org/10.1016/j.socscimed.2005.03.014>. [PubMed: 15922499]
97. van der Weijden T, van Velsen M, Dinant GJ, van Hasselt CM, Grol R. Unexplained complaints in general practice: prevalence, patients' expectations, and professionals' test-ordering behavior. *Med Decis Making.* 2003; 23(3):226–231. <http://dx.doi.org/10.1177/0272989X03023003004>. [PubMed: 12809320]
98. Ali A, Vitulano L, Lee R, Weiss TR, Colson ER. Experiences of patients identifying with chronic Lyme disease in the healthcare system: a qualitative study. *BMC Fam Pract.* 2014; 15:79. <http://dx.doi.org/10.1186/1471-2296-15-79>. [PubMed: 24885888]
99. Hatcher S, Arroll B. Assessment and management of medically unexplained symptoms. *BMJ.* 2008; 336(7653):1124–1128. <http://dx.doi.org/10.1136/bmj.39554.592014.BE>. [PubMed: 18483055]
100. McEwen BS. Physiology and neurobiology of stress and adaptation: central role of the brain. *Physiol Rev.* 2007; 87(3):873–904. <http://dx.doi.org/10.1152/physrev.00041.2006>. [PubMed: 17615391]
101. Smith RC, Lein C, Collins C, et al. Treating patients with medically unexplained symptoms in primary care. *J Gen Intern Med.* 2003; 18(6):478–489. <http://dx.doi.org/10.1046/j.1525-1497.2003.20815.x>. [PubMed: 12823656]
102. Salmon P, Peters S, Clifford R, et al. Why do general practitioners decline training to improve management of medically unexplained symptoms? *J Gen Intern Med.* 2007; 22(5):565–571. <http://dx.doi.org/10.1007/s11606-006-0094-z>. [PubMed: 17443362]
103. Nordin TA, Hartz AJ, Noyes R Jr. et al. Empirically identified goals for the management of unexplained symptoms. *Fam Med.* 2006; 38(7):476–482. [PubMed: 16823672]
104. Piro-Boisset M, Esdaile JM, Fitzcharles MA. Alternative medicine use in fibromyalgia syndrome. *Arthritis Care Res.* 1996; 9(1):13–17. <http://dx.doi.org/10.1002/art.1790090105>. [PubMed: 8945108]
105. Paterson C, Taylor RS, Griffiths P, et al. Acupuncture for 'frequent attenders' with medically unexplained symptoms: a randomised controlled trial (CACTUS study). *Br J Gen Pract.* 2011; 61(587):e295–305. <http://dx.doi.org/10.3399/bjgp11X572689>. [PubMed: 21801508]
106. Dimmock S, Troughton P, Bird H. Factors predisposing to the resort of complementary therapies in patients with fibromyalgia. *Clin Rheumatol.* 1996; 15(5):478–482. <http://dx.doi.org/10.1007/BF02229645>. [PubMed: 8894361]

107. Sarac AJ, Gur A. Complementary and alternative medical therapies in fibromyalgia. *Curr Pharm Des.* 2006; 12(1):47–57. <http://dx.doi.org/10.2174/138161206775193262>. [PubMed: 16454724]
108. Jones JF, Maloney EM, Boneva RS, Jones AB, Reeves WC. Complementary and alternative medical therapy utilization by people with chronic fatiguing illnesses in the United States. *BMC Complement Altern Med.* 2007; 7:12. <http://dx.doi.org/10.1186/1472-6882-7-12>. [PubMed: 17459162]
109. Ali A, McCarthy PL. Complementary and integrative methods in fibromyalgia. *Pediatr Rev.* 2014; 35(12):510–518. <http://dx.doi.org/10.1542/pir.35-12-510>. [PubMed: 25452660]
110. Mist SD, Firestone KA, Jones KD. Complementary and alternative exercise for fibromyalgia: a meta-analysis. *J Pain Res.* 2013; 6:247–260. <http://dx.doi.org/10.2147/JPR.S32297>. [PubMed: 23569397]
111. Goyal M, Singh S, Sibinga EM, et al. Meditation programs for psychological stress and well-being: a systematic review and meta-analysis. *JAMA Intern Med.* 2014; 174(3):357–368. <http://dx.doi.org/10.1001/jamainternmed.2013.13018>. [PubMed: 24395196]
112. Hsu MC, Schubiner H, Lumley MA, Stracks JS, Clauw DJ, Williams DA. Sustained pain reduction through affective self-awareness in fibromyalgia: a randomized controlled trial. *J Gen Intern Med.* 2010; 25(10):1064–1070. <http://dx.doi.org/10.1007/s11606-010-1418-6>. [PubMed: 20532650]
113. Morone NE, Greco CM. Mind-body interventions for chronic pain in older adults: a structured review. *Pain Med.* 2007; 8(4):359–375. <http://dx.doi.org/10.1111/j.1526-4637.2007.00312.x>. [PubMed: 17610459]
114. Wang C, Schmid CH, Rones R, et al. A randomized trial of tai chi for fibromyalgia. *N Engl J Med.* 2010; 363(8):743–754. <http://dx.doi.org/10.1056/NEJMoa0912611>. [PubMed: 20818876]
115. Ali A, Njike VY, Northrup V, et al. Intravenous micronutrient therapy (Myers' Cocktail) for fibromyalgia: a placebo-controlled pilot study. *J Altern Complement Med.* 2009; 15(3):247–257. <http://dx.doi.org/10.1089/acm.2008.0410>. [PubMed: 19250003]
116. Little P, Lewith G, Webley F, et al. Randomised controlled trial of Alexander technique lessons, exercise, and massage (ATEAM) for chronic and recurrent back pain. *BMJ.* 2008; 337:a884. <http://dx.doi.org/10.1136/bmj.a884>. [PubMed: 18713809]
117. Abrams DI, Dolor R, Roberts R, et al. The BraveNet prospective observational study on integrative medicine treatment approaches for pain. *BMC Complement Altern Med.* 2013; 13:146. <http://dx.doi.org/10.1186/1472-6882-13-146>. [PubMed: 23800144]
118. Ritenbaugh C, Hammerschlag R, Calabrese C, et al. A pilot whole systems clinical trial of traditional Chinese medicine and naturopathic medicine for the treatment of temporomandibular disorders. *J Altern Complement Med.* 2008; 14(5):475–487. <http://dx.doi.org/10.1089/acm.2007.0738>. [PubMed: 18564953]
119. Sundberg T, Petzold M, Wandell P, Ryden A, Falkenberg T. Exploring integrative medicine for back and neck pain - a pragmatic randomised clinical pilot trial. *BMC Complement Altern Med.* 2009; 9:33. <http://dx.doi.org/10.1186/1472-6882-9-33>. [PubMed: 19735542]
120. Herman PM, Szczurko O, Cooley K, Seely D. A naturopathic approach to the prevention of cardiovascular disease: cost-effectiveness analysis of a pragmatic multi-worksites randomized clinical trial. *J Occup Environ Med.* 2014; 56(2):171–176. <http://dx.doi.org/10.1097/JOM.000000000000066>. [PubMed: 24451612]
121. Schneiderman N, Ironson G, Siegel SD. Stress and health: psychological, behavioral, and biological determinants. *Annu Rev Clin Psychol.* 2005; 1:607–628. <http://dx.doi.org/10.1146/annurev.clinpsy.1.102803.144141>. [PubMed: 17716101]
122. Ryff CD, Singer BH, Dienberg Love G. Positive health: connecting well-being with biology. *Philos Trans R Soc Lond B Biol Sci.* 2004; 359(1449):1383–1394. <http://dx.doi.org/10.1098/rstb.2004.1521>. [PubMed: 15347530]
123. Kok BE, Coffey KA, Cohn MA, et al. How positive emotions build physical health: perceived positive social connections account for the upward spiral between positive emotions and vagal tone. *Psychol Sci.* 2013; 24(7):1123–1132. <http://dx.doi.org/10.1177/0956797612470827>. [PubMed: 23649562]

124. Giltay EJ, Geleijnse JM, Zitman FG, Hoekstra T, Schouten EG. Dispositional optimism and all-cause and cardiovascular mortality in a prospective cohort of elderly dutch men and women. *Arch Gen Psychiatry*. 2004; 61(11):1126–1135. <http://dx.doi.org/10.1001/archpsyc.61.11.1126>. [PubMed: 15520360]
125. Kobasa SC. Stressful life events, personality, and health: an inquiry into hardiness. *J Pers Soc Psychol*. 1979; 37(1):1–11. <http://dx.doi.org/10.1037/0022-3514.37.1.1>. [PubMed: 458548]
126. Miller FG, Colloca L, Kaptchuk TJ. The placebo effect: illness and interpersonal healing. *Perspect Biol Med*. 2009; 52(4):518–539. <http://dx.doi.org/10.1353/pbm.0.0115>. [PubMed: 19855122]
127. Hauser W, Sarzi-Puttini P, Tolle TR, Wolfe F. Placebo and nocebo responses in randomised controlled trials of drugs applying for approval for fibromyalgia syndrome treatment: systematic review and meta-analysis. *Clin Exp Rheumatol*. 2012; 30(6 Suppl 74):78–87. [PubMed: 23137770]
128. Finniss DG, Kaptchuk TJ, Miller F, Benedetti F. Biological, clinical, and ethical advances of placebo effects. *Lancet*. 2010; 375(9715):686–695. [http://dx.doi.org/10.1016/S0140-6736\(09\)61706-2](http://dx.doi.org/10.1016/S0140-6736(09)61706-2). [PubMed: 20171404]
129. Cohen MH, Kemper KJ, Stevens L, Hashimoto D, Gilmour J. Pediatric use of complementary therapies: ethical and policy choices. *Pediatrics*. 2005; 116(4):e568–575. <http://dx.doi.org/10.1542/peds.2005-0496>. [PubMed: 16199686]
130. Kam-Hansen S, Jakubowski M, Kelley JM, et al. Altered placebo and drug labeling changes the outcome of episodic migraine attacks. *Sci Transl Med*. 2014; 6(218):218ra5. <http://dx.doi.org/10.1126/scitranslmed.3006175>.
131. Kaptchuk TJ, Miller FG. Placebo Effects in Medicine. *N Engl J Med*. 2015; 373(1):8–9. <http://dx.doi.org/10.1056/NEJMp1504023>. [PubMed: 26132938]
132. Tilburt JC, Emanuel EJ, Kaptchuk TJ, Curlin FA, Miller FG. Prescribing "placebo treatments": results of national survey of US internists and rheumatologists. *BMJ*. 2008; 337:a1938. [PubMed: 18948346]
133. National Center for Complementary and Integrative Health. Complementary, Alternative, or Integrative Health: What's In a Name?. 2014.
134. Roter DL, Hall JA, Kern DE, Barker LR, Cole KA, Roca RP. Improving physicians' interviewing skills and reducing patients' emotional distress. A randomized clinical trial. *Arch Intern Med*. 1995; 155(17):1877–1884. <http://dx.doi.org/10.1001/archinte.1995.00430170071009>. [PubMed: 7677554]
135. Kligler B, Maizes V, Schachter S, et al. Core competencies in integrative medicine for medical school curricula: a proposal. *Acad Med*. 2004; 79(6):521–531. <http://dx.doi.org/10.1097/00001888-200406000-00006>. [PubMed: 15165971]
136. Abbo ED, Zhang Q, Zelder M, Huang ES. The Increasing Number of Clinical Items Addressed During the Time of Adult Primary Care Visits. *J Gen Intern Med*. 2008; 23(12):2058–2065. <http://dx.doi.org/10.1007/s11606-008-0805-8>. [PubMed: 18830762]
137. Teutsch C. Patient-doctor communication. *Med Clin North Am*. 2003; 87(5):1115–1145. [http://dx.doi.org/10.1016/S0025-7125\(03\)00066-X](http://dx.doi.org/10.1016/S0025-7125(03)00066-X). [PubMed: 14621334]
138. Cohen MH, Eisenberg DM. Potential physician malpractice liability associated with complementary and integrative medical therapies. *Ann Intern Med*. 2002; 136(8):596–603. <http://dx.doi.org/10.7326/0003-4819-136-8-200204160-00009>. [PubMed: 11955028]
139. Deng, G.; Weber, W.; Sood, A.; Kemper, K. Integrative Medicine Research: Context and Priorities; Commissioned paper for IOM of the National Academies. Summit on Integrative Medicine and the Health of the Public; 2009.
140. Barrett B, Marchand L, Scheder J, et al. What complementary and alternative medicine practitioners say about health and health care. *Ann Fam Med*. 2004; 2(3):253–259. <http://dx.doi.org/10.1370/afm.81>. [PubMed: 15209203]
141. Barrett B, Marchand L, Scheder J, et al. Themes of holism, empowerment, access, and legitimacy define complementary, alternative, and integrative medicine in relation to conventional biomedicine. *J Altern Complement Med*. 2003; 9(6):937–947. <http://dx.doi.org/10.1089/107555303771952271>. [PubMed: 14736364]

142. McGinnis JM, Williams-Russo P, Knickman JR. The case for more active policy attention to health promotion. *Health Aff (Millwood)*. 2002; 21(2):78–93. <http://dx.doi.org/10.1377/hlthaff.21.2.78>. [PubMed: 11900188]
143. Hu P, Reuben DB. Effects of managed care on the length of time that elderly patients spend with physicians during ambulatory visits: National Ambulatory Medical Care Survey. *Med Care*. 2002; 40(7):606–613. <http://dx.doi.org/10.1097/00005650-200207000-00007>. [PubMed: 12142776]
144. Freeman JD, Kadiyala S, Bell JF, Martin DP. The causal effect of health insurance on utilization and outcomes in adults: a systematic review of U.S. studies. *Med Care*. 2008; 46(10):1023–1032. <http://dx.doi.org/10.1097/MLR.0b013e318185c913>. [PubMed: 18815523]
145. Burwell SM. Setting value-based payment goals--HHS efforts to improve U.S. health care. *N Engl J Med*. 2015; 372(10):897–899. <http://dx.doi.org/10.1056/NEJMp1500445>. [PubMed: 25622024]
146. Fritts M, Calvo A, Jonas W, Bezold C. Integrative medicine and health disparities: a scoping meeting. *Explore (NY)*. 2009; 5(4):228–241. <http://dx.doi.org/10.1016/j.explore.2009.05.013>. [PubMed: 19608112]
147. GBD 2013 Mortality and Causes of Death Collaborators. Global, regional, and national age-sex specific all-cause and cause-specific mortality for 240 causes of death, 1990–2013: a systematic analysis for the Global Burden of Disease Study 2013. *Lancet*. 2015; 385(9963):117–171. [http://dx.doi.org/10.1016/S0140-6736\(14\)j-f-2](http://dx.doi.org/10.1016/S0140-6736(14)j-f-2). [PubMed: 25530442]
148. Lexchin J, Bero LA, Djulbegovic B, Clark O. Pharmaceutical industry sponsorship and research outcome and quality: systematic review. *BMJ*. 2003; 326(7400):1167–1170. <http://dx.doi.org/10.1136/bmj.326.7400.1167>. [PubMed: 12775614]
149. Bes-Rastrollo M, Schulze MB, Ruiz-Canela M, Martinez-Gonzalez MA. Financial conflicts of interest and reporting bias regarding the association between sugar-sweetened beverages and weight gain: a systematic review of systematic reviews. *PLoS Med*. 2013; 10(12):e1001578. discussion e1001578. [PubMed: 24391479]
150. Ma B, Guo J, Qi G, et al. Epidemiology, quality and reporting characteristics of systematic reviews of traditional Chinese medicine interventions published in Chinese journals. *PLoS One*. 2011; 6(5):e20185. <http://dx.doi.org/10.1371/journal.pone.0020185>. [PubMed: 21633698]
151. Singh, S.; Ernst, E. *Trick or treatment : the undeniable facts about alternative medicine*. 1st American. W.W. Norton; New York: 2008.
152. Cohen, M. Key Points for Physicians and other Healthcare Licensees to Consider When Selling Dietary Supplements to Patients. <http://michaelhcohen.com/2014/12/key-points-for-physicians-and-other-healthcare-licensees-to-consider-when-selling-dietary-supplements-to-patients/>
153. Health Resources and Services Administration. Integrative Medicine Program (IMP) Announcement Number: HRSA-12-180. 2012.
154. American College of Preventive Medicine. What is Preventive Medicine?. 2015. www.acpm.org/?page=whatispm
155. Moghimi Y. The "PharmFree" campaign: educating medical students about industry influence. *PLoS Med*. 2006; 3(1):e30. <http://dx.doi.org/10.1371/journal.pmed.0030030>. [PubMed: 16435890]
156. Steinbrook R, Ross JS. "Transparency reports" on industry payments to physicians and teaching hospitals. *JAMA*. 2012; 307(10):1029–1030. <http://dx.doi.org/10.1001/jama.2012.211>. [PubMed: 22336624]
157. Colla CH, Morden NE, Sequist TD, Schpero WL, Rosenthal MB. Choosing wisely: prevalence and correlates of low-value health care services in the United States. *J Gen Intern Med*. 2015; 30(2):221–228. <http://dx.doi.org/10.1007/s11606-014-3070-z>. [PubMed: 25373832]
158. Whitsitt V, Beehner C, Welch C. The role of good manufacturing practices for preventing dietary supplement adulteration. *Anal Bioanal Chem*. 2013; 405(13):4353–4358. <http://dx.doi.org/10.1007/s00216-012-6663-6>. [PubMed: 23361225]
159. Benedetti F. Placebo effects: from the neurobiological paradigm to translational implications. *Neuron*. 2014; 84(3):623–637. <http://dx.doi.org/10.1016/j.neuron.2014.10.023>. [PubMed: 25442940]

160. Jonas WB. Reframing placebo in research and practice. *Philos Trans R Soc Lond B Biol Sci*. 2011; 366(1572):1896–1904. <http://dx.doi.org/10.1098/rstb.2010.0405>. [PubMed: 21576147]
161. National Center for Complementary and Alternative Medicine. Expanding Horizons of Health Care: Strategic Plan 2005-2009. National Institutes of Health; Bethesda, MD: 2005.
162. Tufts Center for the Study of Drug Development. Cost to Develop and Win Marketing Approval for a New Drug Is \$2.6 Billion. 2014. http://csdd.tufts.edu/news/complete_story/pr_tufts_csdd_2014_cost_study
163. Verhoef MJ, Lewith G, Ritenbaugh C, Boon H, Fleishman S, Leis A. Complementary and alternative medicine whole systems research: beyond identification of inadequacies of the RCT. *Complement Ther Med*. 2005; 13(3):206–212. <http://dx.doi.org/10.1016/j.ctim.2005.05.001>. [PubMed: 16150375]
164. Wayne PM, Kaptchuk TJ. Challenges inherent to t'ai chi research: part I--t'ai chi as a complex multicomponent intervention. *J Altern Complement Med*. 2008; 14(1):95–102. <http://dx.doi.org/10.1089/acm.2007.7170A>. [PubMed: 18199021]
165. Hawk C, Khorsan R, Lisi AJ, Ferrance RJ, Evans MW. Chiropractic care for nonmusculoskeletal conditions: a systematic review with implications for whole systems research. *J Altern Complement Med*. 2007; 13(5):491–512. <http://dx.doi.org/10.1089/acm.2007.7088>. [PubMed: 17604553]
166. Edman JS, Roberts RS, Dusek JA, Dolor R, Wolever RQ, Abrams DI. Characteristics of cancer patients presenting to an integrative medicine practice-based research network. *Integr Cancer Ther*. 2014; 13(5):405–410. <http://dx.doi.org/10.1177/1534735414537876>. [PubMed: 24913179]
167. Ho DV, Nguyen J, Liu MA, Nguyen AL, Kilgore DB. Use of and interests in complementary and alternative medicine by Hispanic patients of a community health center. *J Am Board Fam Med*. 2015; 28(2):175–183. <http://dx.doi.org/10.3122/jabfm.2015.02.140210>. [PubMed: 25748757]
168. Gardiner P, Dresner D, Barnett KG, Sadikova E, Saper R. Medical group visits: a feasibility study to manage patients with chronic pain in an underserved urban clinic. *Glob Adv Health Med*. 2014; 3(4):20–26. <http://dx.doi.org/10.7453/gahmj.2014.011>. [PubMed: 25105072]
169. McKee MD, Kligler B, Fletcher J, et al. Outcomes of acupuncture for chronic pain in urban primary care. *J Am Board Fam Med*. 2013; 26(6):692–700. <http://dx.doi.org/10.3122/jabfm.2013.06.130003>. [PubMed: 24204065]
170. Office of the Legislative Counsel. Compilation of Patient Protection and Affordable Care Act: Extracted sections concerning Patient-Centered Outcomes Research and the Authorization of the Patient-Centered Outcomes Research Institute (PCORI). 2010. Session tCd
171. Patient-Centered Outcomes Research Institute. Acupuncture Approaches to Decrease Disparities in Outcomes of Pain Treatment-A Two Arm Comparative Effectiveness Trial (AADOPT-2). 2014. www.pcori.org/research-results/2014/acupuncture-approaches-decrease-disparities-outcomes-pain-treatment-two-arm
172. Patient-Centered Outcomes Research Institute. Integrative Medicine Group Visits: A Patient-Centered Approach to Reducing Chronic Pain and Depression in a Disparate Urban Population. 2013. www.pcori.org/research-results/2013/integrative-medicine-group-visits-patient-centered-approach-reducing-chronic
173. Center for Optimal Integration. Project for Integrative Health and the Triple Aim (PIHTA). 2013. <http://optimalintegration.org/project-pihta/pihta.php>
174. Mao JJ, Cohen L. Advancing the science of integrative oncology to inform patient-centered care for cancer survivors. *J Natl Cancer Inst Monogr*. 2014; 2014(50):283–284. <http://dx.doi.org/10.1093/jncimonographs/igu038>. [PubMed: 25749588]

Table 1Benefit and Risk Ratio and Selection of Therapies. From ¹³⁹

		EFFECTIVE	
		YES	NO
SAFE	YES	Use	Tolerate
	NO	Monitor	Avoid

Table 2

The Clinical Applications of Research Evidence Construct. Adapted and expanded from³¹

Safety	Efficacy	Science	Other therapeutic options	Patient preference	Cost / Accessibility	Utilization frequency of treatment in question
High	High	Decisive	None that is superior	Prefers recommended approach	Not a concern	Always
Probable	Possible	Unclear	None / few	Anything that will work	Needs consideration	Often
Low	Low	Absent / opposed	Many that are superior	Anything that will work	Prohibitive	Never